

$$CP = cp * (\frac{Q}{2} + ss) \quad C_e = c_e \cdot (\frac{D}{Q})$$

$$Ca = ca \cdot D$$

$$Q^* = \sqrt{\frac{2 \cdot c_e \cdot D}{c_p}} \equiv \text{E.O.Q.}$$

$$Tr = \frac{t}{f}$$

$$f = \frac{D}{Q}$$

Punto de Pedido = Tiempo de suministro x Demanda diaria (D/t) + ss

$$Pp = (Ts - E \times Tr) \times D / t + ss$$

Model with Planned Shortages

$$TC = K \frac{D}{Q} + h \frac{(Q-S)^2}{2Q} + B \frac{S^2}{2Q}$$

$$Q^* = \sqrt{\frac{2DK}{h} \left(\frac{h+B}{B} \right)}$$

$$S^* = Q^* \left(\frac{h}{h+B} \right)$$

B: shortage (stockout) cost

t1: can we supply

$$t1 = (Q^* - S^*) / D \quad t2 = S^* / D$$

t2: not supply a demand